通原第三章作业1

吴海迪 班级2020213107 学号2020213248

3.1 解：

(1) ∵ $E[Y(t)] = E[X(t)]E[\cos(2\pi f_c t+\theta)]$

$= 0$

∴ $Y(t)$ 均值为0。

$R_Y(\tau) = E[Y(t+\tau)Y(t)]$

$= E[X(t+\tau)\cos(2\pi f_c t+\theta+2\pi f_c\tau)\cdot X(t)\cos(2\pi f_c t+\theta)]$

$= E[X(t)X(t+\tau)]\cdot E[\cos(2\pi f_c t+\theta)\cos(2\pi f_c t+\theta+2\pi f_c\tau)]$

$= R_X(\tau)\cdot\frac{1}{2}E[\cos 2\pi f_c\tau+\cos(4\pi f_c t+2\theta+2\pi f_c\tau)]$

$= \frac{1}{2}R_X(\tau)\cos(2\pi f_c\tau)+\frac{1}{2}R_X(\tau)E[\cos(4\pi f_c t+2\theta+2\pi f_c\tau)]$

∵ θ 在 $(0,2\pi)$ 均匀分布

∴ $P(\theta)=\frac{1}{2\pi}$ 当 $\theta\in[0,2\pi)$

∴ $E[\cos(4\pi f_c t+2\theta+2\pi f_c\tau)]=0$

∴ $R_Y(\tau)=\frac{1}{2}R_X(\tau)\cos(2\pi f_c\tau)$

∵ 功率谱密度 $P_Y(f)$ 与自相关函数 $R_Y(\tau)$ 为傅式变换对

∴ $P_Y(f)=\int_{-\infty}^{+\infty}R_Y(\tau)\cdot e^{-j2\pi f\tau}d\tau$

$= \frac{1}{2}\int_{-\infty}^{+\infty}R_X(\tau)\cos 2\pi f_c\tau\cdot e^{-j2\pi f\tau}d\tau$

$= \frac{1}{4}[P_X(f-f_c)+P_X(f+f_c)]$

(2) ∵ $E[Y(t)] = E[X(t)]E[\cos(2\pi f_c t+\theta)]$

$= 0$

∴ $Y(t)$ 均值为0.

∵ 同理

$R_Y(\tau)=\frac{1}{2}R_X(\tau)\cos(2\pi f_c\tau)+\frac{1}{2}R_X(\tau)E[\cos(4\pi f_c t+2\theta+2\pi f_c\tau)]$

且θ为常数

∴ $R_Y(\tau)=\frac{1}{2}R_X(\tau)\cos(2\pi f_c\tau)+\frac{1}{2}R_X(\tau)\cos(4\pi f_c t+2\theta+2\pi f_c\tau)$

$Y(t)$为循环平稳过程

∵ $\overline{\cos(4\pi f_c t+2\theta+2\pi f_c\tau)}=0$ (取时间平均)

∴ $\overline{R_Y(\tau)}=\frac{1}{2}R_X(\tau)\cos 2\pi f_c\tau$

⇒同理, $R_Y(f)=\frac{1}{4}[P_X(f-f_c)+P_X(f+f_c)]$

3.3 解:

∵ 通过线性系统的平稳过程仍为平稳过程

∴ $Y(t)$为窄带过程, 带宽为B

∵ $P_Y(f)=\frac{N_0}{2}|H(f)|^2|_{f=2\pi f}$

$=2N_0(\pi f)^2$ 当 $|f\pm f_c|\le\frac{B}{2}$

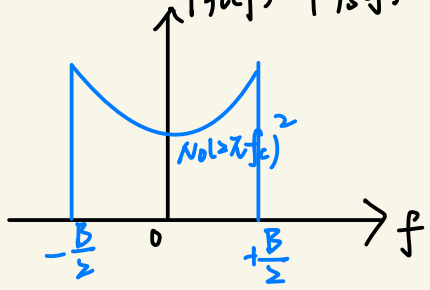
$P_{Yc}(f)=P_{Ys}(f)=P_Y(f+f_c)+P_Y(f-f_c)$ 当 $|f|\le\frac{B}{2}$

∴ $P_{Yc}(f)=P_{Ys}(f)$

$=\begin{cases}2N_0\pi^2(f+f_c)^2+2N_0\pi^2(f-f_c)^2, & |f|\le\frac{B}{2}\\ 0 & \text{其他}\end{cases}$

$=\begin{cases}4N_0\pi^2(f^2+f_c^2), & |f|\le\frac{B}{2}\\ 0 & \text{其他}\end{cases}$

图:



3.5 解:

∵ u(t) 是 $A\cos 2\pi f_c t$ 对应的输出

∴ $u(t) = [A\cos 2\pi f_c t \times \cos(2\pi f_c t + \theta)]_{lowpass}$

$= \frac{A}{2}\cos\theta$

∵ v(t) 是与 x(t) 对应的输出

$v(t) = [(x_c(t)\cos 2\pi f_c t - x_s(t)\sin 2\pi f_c t) \times \cos(2\pi f_c t + \theta)]_{lowpass}$

$= \frac{1}{2}x_c(t)\cos\theta + \frac{1}{2}x_s(t)\sin\theta$

$\Rightarrow P_u = \frac{A^2\cos^2\theta}{4}$ $P_v = \cos^2\theta \cdot \frac{1}{4}\sigma_x^2 + \sin^2\theta \cdot \frac{1}{4}\sigma_x^2 = \frac{\sigma_x^2}{4}$

(1) 当 θ 为常数.

$\frac{P_u}{P_v} = \frac{A^2\cos^2\theta}{\sigma_x^2}$

(2) $P_u = \frac{A^2\cos^2\theta}{4}$ $\Rightarrow$ 平均功率 $= E_\theta[\frac{A^2\cos^2\theta}{4}]$

$= \frac{A^2}{4}\int_{-\infty}^{+\infty}\frac{\cos^2\theta}{\sqrt{2\pi}\sigma^2}e^{-\frac{\theta^2}{2\sigma^2}}d\theta$

$\therefore \frac{P_u}{P_v} = \frac{A^2}{\sigma_x^2}\cdot E[\cos^2\theta]$

$= \frac{A^2}{2\sigma_x^2}\cdot E[1+\cos 2\theta]$

$= \frac{A^2}{2\sigma_x^2}[1+\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi}\sigma^2}e^{\frac{-\theta^2}{2\sigma^2}}\cos 2\theta d\theta]$

$= \frac{A^2}{2\sigma_x^2}(1+e^{-2\sigma^2})$

3.6 解:

(1) $P_y(f) = \frac{N_0}{2}|j2\pi f|^2 = 2\pi^2 N f^2 = 3.95\times 10^{-5} f^2$ W/Hz

(2) $P_{y_0} = \int_{-B}^{B} P_y(f)\,df = 2\int_0^B 2\pi^2 N_0 f^2\,df$

$= \frac{4\pi^2 N_0 B^3}{3}$

$= 0.0263\,W$